PAPER ID-311178

Roll No: | | | | | | | | | | | | | |

BTECH
(SEM V) THEORY EXAMINATION 2024-25
AUTOMOTIVE CHASSIS AND SUSPENSION

TIME: 3 HRS **M.MARKS: 70**

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt *all* questions in brief. **2 x 07 = 14**

Q no.	Question	CO	Level
a.	Discuss the various loads acting on the frame.	1	K2
b.	Define Chassis. Name its different components	1	K1
c.	Describe the different part of differential gear.	2	K2
d.	Discuss the characteristics of universal joints used in vehicle.	2	K2
e.	Describe the function of master cylinder in braking system	3	K2
f.	Explain the significance of camber and castor in steering geometry	4	K2
g.	Describe the working of hill assist system.	5	K2

SECTION B

2. Attempt any *three* of the following: **07 x 3 = 07**

Q no.	Question	CO	Level
a.	Explain the characteristics of front engine front wheel drive along with its advantages and disadvantages.	1	K2
b.	Describe the functioning of synchromesh gear box.	2	K2
c.	Discuss the working of air brake system in automobile.	3	K2
d.	Derive the condition of true rolling condition for true rolling of wheels	4	K3
e.	Explain in detail the construction of tyre of automobile.	5	K2

SECTION C

3. Attempt any *one* part of the following: **07 x 1 = 07**

Q no.	Question	CO	Level
a.	Discuss the characteristics of ladder frame along with its advantages and disadvantages	1	K2
b.	Describe the characteristics of front engine rear wheel drive along with its advantages and disadvantages	1	K2

4. Attempt any *one* part of the following: **07 x 1 = 07**

Q no.	Question	CO	Level
a.	The design is required to be made of a sliding mesh gear box with approximate gear ratios in forward speeds as 1.0, 1.5 and 2.5 and in reverse speed as 3.2. The diametral pitch of each gear is 4 mm and the smallest pinion is to have at least 12 teeth. The centre distance between the layshaft and main shaft is 84 mm. Evaluate the suitable number of teeth on different gears and exact gear ratio's thus available	2	K4
b.	Explain the working of constant mesh gearbox in automobile.	2	K2

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5. Attempt any *one* part of the following: 07 x 1 = 07

Q no.	Question	CO	Level
a.	A vehicle weighing 13500 N has a wheel base of 2.6 m. The centre of gravity lies 1.3 m behind the front axle and 0.6 m above the ground. The vehicle moves on the level ground with a speed of 54 km/hr. When brakes are applied to it on all four wheels, the car stop in a distance of 25 m. Calculate the coefficient of friction. With the road conditions remaining same, the vehicle descends a hill of 1 in 10 gradient and is braked on front wheels only. Calculate the load distribution and stopping distance of the vehicle.	3	K3
b.	Explain the working of Macpherson strut used in automobile.	3	K2

6. Attempt any *one* part of the following: 07 x 1 = 07

Q no.	Question	CO	Level
a.	Derive the condition for davis steering mechanism	4	K3
b.	A vehicle using Ackermann steering system has a wheel base of 280 cm and front wheel track 122 cm, the distance between kingpin axis is108cm. If the maximum deflection of inner front wheel is 45°. Calculate the turning radius of outer front wheel	4	K3

7. Attempt any *one* part of the following: 07 x 1 = 07

Q no.	Question	CO	Level
a.	Explain the different types of bearings used for radial loads.	5	K2
b.	Explain the working of ABS system used in automobiles.	5	K2

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